Silverio Martínez Andrés Grupo 3S Tarea 25

Nombre:

Tema:

Día Mes Año Folio

A | 10

Tarea 25

1.- Derivar las siguientes funciones

a) $y = \ln 2x$

$y' = \frac{\cancel{2}}{\cancel{2}x} = \frac{1}{x}$

b) $f(x) = \ln(x^2+1)$

$f'(x) = \frac{2x}{(x^2+1)}$

c) $g(x) = x \ln x$

$g'(x) = \cancel{x}\frac{1}{\cancel{x}} + \ln x\,(1)$

$g'(x) = 1 + \ln x$

d) $h(x) = (\ln x)^3$

$h'(x) = 3(\ln x)^2 \boxed{\frac{1}{x}}$ RC

$h'(x) = \frac{3(\ln x)^2}{x}$

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e) $y = \ln\sqrt{x+1} = \ln(x+1)^{1/2} = \frac{1}{2}\ln(x+1)$

$y' = \frac{1}{2}\cdot\frac{1}{x+1} = \frac{1}{2(x+1)} = \frac{1}{2x+2}$

f) $f(x) = \ln\frac{x(x^2+1)^2}{\sqrt{2x^3-1}} = \ln x(x^2+1)^2 - \ln\sqrt{2x^3-1}$

$f(x) = \ln x + \ln(x^2+1)^2 - \ln(2x^3-1)^{1/2}$

$f(x) = \ln x + 2\ln(x^2+1) - \frac{1}{2}\ln(2x^3-1)$

$f'(x) = \frac{1}{x} + 2\frac{2x}{x^2+1} - \frac{1}{2}\frac{6x^2}{2x^3-1}$

$f'(x) = \frac{1}{x} + \frac{4x}{x^2+1} - \frac{3x^2}{2x^3-1}$

g) $y = \ln|\cos x|$

$y' = \frac{-\operatorname{sen} x}{\cos x} = -\operatorname{tg} x$

h) $f(x) = \ln|\operatorname{sen} x|$

$f'(x) = \frac{\cos(-x)}{\operatorname{sen} x} = \cot x$

Escaneado con CamScanner

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2: Utilizando derivacion logaritmica, obtener la derivada de

$$y = \frac{(x-2)^2}{\sqrt{x^2+1}}$$

$$\ln y = \ln \frac{(x-2)^2}{\sqrt{x^2+1}}$$

$$\ln y = \ln (x-2)^2 - \ln \sqrt{x^2+1}$$

$$\ln y = 2 \ln (x-2) - \ln (x^2+1)^{1/2}$$

$$\ln y = 2 \ln (x-2) - \frac{1}{2} \ln (x^2+1)$$

$$\frac{y'}{y} = 2 \frac{1}{x-2} - \frac{1}{\cancel{2}} \frac{\cancel{2}x}{x^2+1}$$

$$\frac{y'}{y} = \frac{2}{x-2} - \frac{x}{x^2+1}$$

$$y' = y \left(\frac{2}{x-2} - \frac{x}{x^2+1} \right)$$

$$y' = \frac{(x-2)^2}{\sqrt{x^2+1}} \left(\frac{2}{x-2} - \frac{x}{x^2+1} \right)$$

$$y' = \frac{(x-2)^2}{\sqrt{x^2+1}} \left(\frac{2(x^2+1) - x(x-2)}{(x-2)(x^2+1)} \right)$$

$$y' = \frac{(x-2)^2}{\sqrt{x^2+1}} \left(\frac{2x^2+2-x^2+2x}{(x-2)(x^2+1)} \right)$$

$$y' = \frac{(x-2)^{\cancel{2}}}{(x^2+1)^{1/2}} \left(\frac{x^2+2x+2}{\cancel{(x-2)}(x^2+1)} \right)$$

$$y' = \frac{(x-2)(x^2+2x+2)}{(x^2+1)^{3/2}}$$